In the K-means algorithm, cluster centroids (μ_k) are updated at each iteration to minimize the sum of squared distances between data points and their assigned centroids. The formula used for this update is:

$$\mu_k = \frac{\sum x_i \mathbb{1}(z_i = k)}{\sum \mathbb{1}(z_i = k)}$$

Here:

- x_i represents i^{th} data point in the dataset.
- z_i represents the cluster assignment for data point x_i, where z_i is an integer indicating the cluster index (k).

Suppose you have a dataset of 9 two-dimensional data points and are performing K-means clustering with $K = 3$ clusters. After the first iteration, clusters C_1, C_2, and C_3 have the following data points:

C_1: {(2, 2), (3, 3), (4,4) }

C_2: { (6, 6), (7, 7),(8,8)}

C_3: {(10, 10), (11, 11), (12, 12)}

After the second iteration, which of the following options represents the updated cluster centroid μ_k for C_1 using the provided formula?

$\mu 1 = (2.5, 2.5)$

$\mu 1 = (4.5, 4.5)$

$\mu 1 = (3, 3)$ ✓

$\mu 1 = (5.5, 5.5)$

$C_1 \Rightarrow (3,3)$

$C_2 \Rightarrow (7,7)$

$C_3 \Rightarrow (11,11)$

$x_i \rightarrow C_1$

$\hookrightarrow C_2$

$$\|x_i - \mu_2\|^2 < \|x_i - \mu_1\|^2$$
$$< \|x_i - \mu_3\|^2$$

6406532240047. ✖ $\mu 1 = (2.5, 2.5)$

6406532240048. ✖ $\mu 1 = (4.5, 4.5)$

6406532240049. ✔ $\mu 1 = (3, 3)$

6406532240050. ✖ $\mu 1 = (5.5, 5.5)$

Consider 9 data points $x_1, x_2, \ldots, x_9$: (2,3),(1,2),(3,1),(5,6),(7,8),(9,9),(6,7),(10,10),(11,11). Suppose we aim to form 3 clusters from these points using the K-Means algorithm. After the first iteration, clusters C_1, C_2, C_3 have the following data points:

C_1: (1, 2), (3, 1)

C_2: (2, 3), (5, 6), (6,7)

C_3: (9, 9), (7, 8), (10, 10), (11, 11)

(2, 3/2)

(13/3, 16/3)

(37/4, 38/4)

(2,3)

$0^2 + (-1.5)^2 = 2.25$

$(2\ldots)^2$ > 4

After the second iteration, to which cluster will the data point (2, 3) be reassigned?

9.25

$\frac{13}{3} - \frac{2 \times 3}{3} = \frac{7}{3}$

C_1

C_2

C_3

It remains in the same cluster as in the first iteration

6406532240051. ✔ C_1

6406532240052. ✖ C_2

6406532240053. ✖ C_3

6406532240054. ✖ It remains in the same cluster as in the first iteration

Which of the following statements are true regarding applying standard PCA on a centered dataset?

A The variance of the dataset along the first principal component is maximum.

B Eigenvectors are always orthogonal to each other in PCA.

C The first principal component is the eigenvector corresponding to the largest eigenvalue of the covariance matrix.

D The first principal component is the eigenvector corresponding to the smallest eigenvalue of the covariance matrix.

6406532240065. ✔ The variance of the dataset along the first principal component is maximum.

6406532240066. ✔ Eigenvectors are always orthogonal to each other in PCA.

6406532240067. ✔ The first principal component is the eigenvector corresponding to the largest eigenvalue of the covariance matrix.

6406532240068. ✖ The first principal component is the eigenvector corresponding to the smallest eigenvalue of the covariance matrix.

Consider a dataset that has 200 data-points, each of which is either a zero or one. The Bernoulli distribution is used to model this data. If the MLE estimate for the parameter p of the Bernoulli distribution is 0.3, how many zeros does the dataset have?

$x_i \rightarrow (1, 0)$

$$\hat{p}_{ml} = \frac{\sum_{i=1}^{n} x_i}{n} \Rightarrow \text{fraction of ones}$$

$1 \Rightarrow 0.3$

$0 \Rightarrow 0.7 \times 200 = 140$

Question Label : Short Answer Question

Consider a dataset that has 200 data-points, each of which is either a zero or one. The Bernoulli distribution is used to model this data. If the MLE estimate for the parameter p of the Bernoulli distribution is 0.3, how many zeros does the dataset have?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

140

Consider a dataset that has 4 points in R^2 that lie on a line passing through the origin:

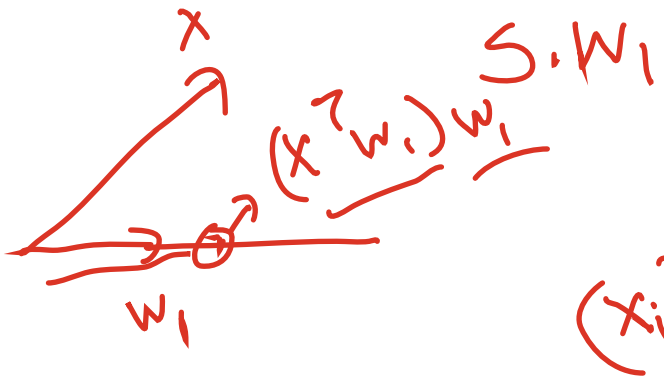
$(x_i^T w_1)^2$ (1000 250 250 1000)

$$D\left\{\begin{bmatrix}-10\\-30\end{bmatrix}, \begin{bmatrix}-5\\-15\end{bmatrix}, \begin{bmatrix}5\\15\end{bmatrix}, \begin{bmatrix}10\\30\end{bmatrix}\right\} \begin{matrix}\Rightarrow x_1\\ \Rightarrow x_2\end{matrix}$$

$x_2 = 3x_1$

If normal PCA is performed on this dataset, what is the variance along the first principal component?

$$\frac{1}{n}\sum_{i=n}^{n}(x_i^T w_1)^2$$



$(x_i^T w_1)^2 \Rightarrow ?$

$$\frac{2500}{4} = \boxed{625}$$

Consider a dataset that has 4 points in R^2 that lie on a line passing through the origin:

$$D\left\{\begin{bmatrix}-10\\-30\end{bmatrix},\begin{bmatrix}-5\\-15\end{bmatrix},\begin{bmatrix}5\\15\end{bmatrix},\begin{bmatrix}10\\30\end{bmatrix}\right\}$$

If normal PCA is performed on this dataset, what is the variance along the first principal component?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

625

$n=50$ $d=5$

Consider a dataset of 50 points all of which lie in $\mathbb{R}^5$. The eigenvalues of the covariance matrix are given below:

$$\boxed{1.7, \quad 1.4, \quad 0.25, \quad 0.20,} \quad 0.005$$

1.70 0
1.400
0 250
0.200
0.005
3.555

If we run the PCA algorithm on this dataset and retain the top-k principal components, what is a good choice of k? (Hint: A good precise mean it captures atleast 95% of variance)

$k=4$

$$\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_d} \geq 0.95 \qquad \frac{\quad}{3.555} \geq 0.95$$

Consider a dataset of 50 points all of which lie in $\mathbb{R}^5$. The eigenvalues of the covariance matrix are given below:

$$1.7, \quad 1.4, \quad 0.25, \quad 0.20, \quad 0.005$$

If we run the PCA algorithm on this dataset and retain the top-k principal components, what is a good choice of k? (Hint: A good precise mean it captures atleast 95% of variance)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

A GMM is fit for a dataset with 5 points. At some time-step in the EM algorithm, the following are the values of λ_k^i for all points in the dataset for the kth mixture after the E-step:

$\lambda_k^1 = 0.3, \lambda_k^2 = 0.1, \lambda_k^3 = 0.5, \lambda_k^4 = 0.5, \lambda_k^5 = 0.3$

What is the estimate of π_k after the M-step? Enter your answer correct to two decimal places.

$$\pi_k = \frac{\sum_{i=1}^{n} \lambda_k^i}{n} = \frac{1.7}{5} = 0.34$$

A GMM is fit for a dataset with 5 points. At some time-step in the EM algorithm, the following are the values of λ_k^i for all points in the dataset for the kth mixture after the E-step:

$\lambda_k^1 = 0.3, \lambda_k^2 = 0.1, \lambda_k^3 = 0.5, \lambda_k^4 = 0.5, \lambda_k^5 = 0.3$

What is the estimate of π_k after the M-step? Enter your answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.3 to 0.4

Suppose you have been given a task of estimating the conversion rate of google click of new online advertisement campaign. You have collected data from a limited sample of 50 users, where 15 of them have converted. You want to use Bayesian estimation with a prior distribution to provide a more robust estimate of the conversion rate.

Assume you have prior information and you decide to use a beta distribution as your prior. You choose a beta distribution with parameters $\alpha = 3$ and $\beta = 4$ to capture your prior beliefs.

Calculate the posterior mean?

$0 \to 1$

$\text{Beta}(\alpha, \beta)$

$\alpha \to \alpha + n_1$, $\beta \to n_0 + \beta$

$3 + 15 = 18$, $35 + 4 = 39$

$1 \quad 0$

$\text{Bernoulli}(p)$

0.315

$$\frac{\alpha}{\alpha+\beta} = \frac{18}{18+39} = \frac{18}{57} = \frac{6}{19}$$

jimmi Course_Team Oct 2023

Suppose you have been given a task of estimating the conversion rate of google click of new online advertisement campaign. You have collected data from a limited sample of 50 users, where 15 of them have converted. You want to use Bayesian estimation with a prior distribution to provide a more robust estimate of the conversion rate.

Assume you have prior information and you decide to use a beta distribution as your prior. You choose a beta distribution with parameters $\alpha = 3$ and $\beta = 4$ to capture your prior beliefs.

Calculate the posterior mean?

Enter your answer correctly up to two decimals

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.2 to 0.3

The original range for this question will be 0.2 to 0.4.

The correct answer is 0.315.

Have a look at this discourse link

https://discourse.onlinedegree.iitm.ac.in/t/quiz-1-answer-key-doubts/108191/11?u=23f2000573

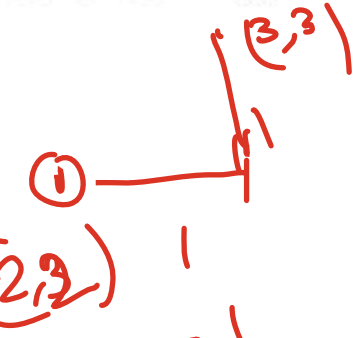
A k-means++ algorithm with k= 3 was applied to the following 2D points:

2 4 16 20 22

(2, 2), (3, 3), (4, 4), (10, 10), (12, 12), (13, 13)

(2, 2) is chosen as the first cluster mean.

$P(\mu_2 = (3,3)) = \frac{2}{64}$



$|2-3| + |2-3|$

Slove

Which point has the highest probability of being chosen as the 2nd cluster mean? Use the Manhattan distance to compute the distances.

(4, 4)

(12, 12)

(10, 10)

(13, 13)

$\# \; ((X-\mu)^2$

$K \; |X-\mu|$

$(3,3) \to \frac{2}{64}$

$(4,4) \to \frac{4}{64}$

$(10,10) \to \frac{16}{4}$

$(12,12) \to \left(\frac{20}{64}\right)$

$(13,13) \to \left(\frac{22}{64}\right)$

Which point has the highest probability of being chosen as the 2nd cluster mean? Use the Manhattan distance to compute the distances.

Options :

6406532240055. ✖ (4, 4)

6406532240056. ✖ (12, 12)

6406532240057. ✔ (10, 10)

6406532240058. ✖ (13, 13)

Correct answer is (13,13). Have a look at the discourse link below

https://discourse.onlinedegree.iitm.ac.in/t/quiz-1-answer-key-doubts/108191/10?u=23f2000573

A k-means++ algorithm with k= 3 was applied to the following 2D points:

(2, 2), (3, 3), (4, 4), (10, 10), (12, 12), (13, 13)

(2, 2) is chosen as the first cluster mean.

Let's assume that (3, 3), (10, 10), and (12, 12) are the cluster means. You introduce a new data point, (−5, 10). Which cluster mean is the data point (−5, 10) closest to when using the Euclidean distance method?

(3, 3) $\rightarrow (3-(-5))^2 + (3-10)^2$ $64+49=113$

(10, 10) $(10-(-5))^2$ $225+$

(12, 12)

None of these

Let's assume that (3, 3), (10, 10), and (12, 12) are the cluster means. You introduce a new data point, (−5, 10). Which cluster mean is the data point (−5, 10) closest to when using the Euclidean distance method?

Options :

6406532240059. ✖ (3, 3)

6406532240060. ✔ (10, 10)

6406532240061. ✖ (12, 12)

6406532240062. ✖ None of these

Correct answer is (3,3). Have a look at the discourse link below

https://discourse.onlinedegree.iitm.ac.in/t/quiz-1-answer-key-doubts/108191/10?u=23f2000573

Suppose that we have a dataset (data matrix) $X \in \mathbb{R}^{d \times n}$, where d denotes the number of features and n denotes the number of samples. Assume that $d >> n$ and the data is centered. The representative line is given by the **unit vector**, pointing in the direction of the line, $\mathbf{w}$.

In all the cases, we maximize the objective function given below

$$f(\mathbf{w}) = \mathbf{w}^T C \mathbf{w}$$

where $C = \frac{1}{n} X X^T$ is a covariance matrix. The kernel matrix is defined to be $K = X^T X$.

$w_1 = X\alpha_1$ $d\times n$, $n\times 1$ $d\times 1$ w_1

Which of the following statements are true?

$d\times d$ $\frac{1}{n}XX^T$ $X^T X$ $n\times n$

The eigenvectors of C and K are related by a scaling factor $r \in \mathbb{R}$ α

Suppose the maximum eigenvalue of C is λ_1, then the maximum eigenvalue of XX^T is $n\lambda_1$

$w_1 \to \lambda_1$ for C

$Cw_1 = \lambda_1 w_1$

The eigenvectors of $C \in \mathbb{R}^d$ and $K \in \mathbb{R}^{d \to n}$ α

$\frac{1}{n} XX^T w_1 = \lambda_1 w_1$

$XX^T w_1 = n\lambda_1 w_1$

Computing eigenvectors using the matrix K reduces number of computations significantly than using the matrix C

$n_a > n_b$

$n_a > n_b$

$n\lambda_1 > n\lambda_2$

The eigenvalues of both C and K are non-negative

≥ 0

$$\frac{1}{n} X X^T w_1 = \lambda_1 w_1 \qquad \Leftrightarrow \lambda_1 \ldots \lambda_d$$

$$X X^T \to n\lambda_1 \ldots n\lambda_d$$

$$X X^T w_1 = (n\lambda_1) w_1$$

$$\lambda_1 \geq \lambda_2$$

$$\underline{n}\lambda_1 \geq \underline{n}\lambda_2$$

6406532240072. ✖ The eigenvectors of C and K are related by a scaling factor $r \in \mathbb{R}$

6406532240073. ✔ Computing eigenvectors using the matrix K reduces number of computations significantly than using the matrix C

6406532240074. ✔ Suppose the maximum eigenvalue of C is λ_1, then the maximum eigenvalue of XX^T is $n\lambda_1$

6406532240075. ✖ The eigenvectors of $C \in \mathbb{R}^d$ and $K \in \mathbb{R}^d$

6406532240076. ✔ The eigenvalues of both C and K are non-negative

Consider the following kernel:

$$k : R^2 \times R^2 \to R$$
$$k(x, y) = (x^T y)^2$$

Which of the following transformation mapping ϕ may correspond to the kernel k?

$\phi([x_1, x_2]^T) = [x_1, x_1 x_2, x_2]^T$

$\phi([x_1, x_2]^T) = [x_1^2, x_1 + x_2, x_2^2]^T$

$\phi([x_1, x_2]^T) = [x_1, \sqrt{2} x_1^2 x_2^2, x_2]^T$

$\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2} x_1 x_2, x_2^2]^T$

$(x_1^2, x_2^2, \sqrt{2} x_1 x_2)^T$

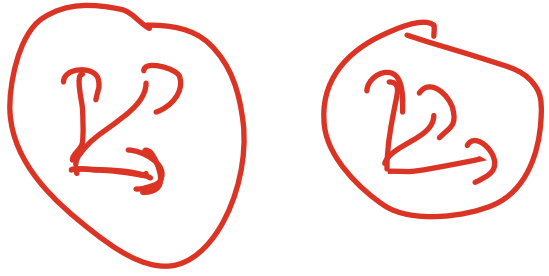
$(y_1^2, y_2^2, \sqrt{2} y_1 y_2)$

$k\left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right] = (x_1 y_1 + x_2 y_2)^2$

$[\ ,\ ,\ ] = x_1^2 y_1^2 + x_2^2 y_2^2 + 2 x_1 x_2 y_1 y_2$

$x_1^2 \to y_1^2$, $x_2^2 \to y_2^2$, $2 x_1 x_2 y_1 y_2 \to \sqrt{2} x_1 x_2 \to \sqrt{2} y_1 y_2$

$$x_1^2 y_1^2 + y_1^2 y_2^2 + 2$$



Consider the following kernel:

$$k : R^2 \times R^2 \to R$$
$$k(x, y) = (x^T y)^2$$

Which of the following transformation mapping ϕ may correspond to the kernel k?

Options :

6406531286416. ✖ $\phi([x_1, x_2]^T) = [x_1, x_1 x_2, x_2]^T$

6406531286417. ✖ $\phi([x_1, x_2]^T) = [x_1^2, x_1 + x_2, x_2^2]^T$

6406531286418. ✖ $\phi([x_1, x_2]^T) = [x_1, \sqrt{2} x_1^2 x_2^2, x_2]^T$

6406531286419. ✔ $\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2} x_1 x_2, x_2^2]^T$

A function k is defined as

$$k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$

$x_2^3 x_1 \neq x_1^3 x_2$

$$k(x_1, x_2) = x_1 x_2 + x_1^2 x_2^2 + x_1^3 x_2 + 1$$

Is k a valid kernel?

$\phi(x)^T \phi(y) \Rightarrow$

Yes

No

A function k is defined as

$$k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$
$$k(x_1, x_2) = x_1 x_2 + x_1^2 x_2^2 + x_1^3 x_2 + 1$$

Is k a valid kernel?

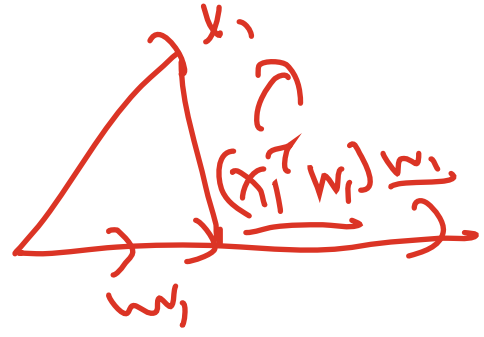
Options :

6406531286420. ✖ Yes

6406531286421. ✔ No

Let X be a data matrix of shape (d, n) for a centered dataset. The first principal component of the dataset is $\left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right]^T$. What will the scalar proxy of the point $[1, 2]^T$ be on the first principal component?

$\hookrightarrow \frac{1}{4} + \frac{3}{4} = \frac{4}{4} = 1$



w_1
$\vdots$
w_k

$(1, 2)\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

$\frac{1}{2} + \frac{2\sqrt{3}}{2} = \frac{1}{2} + \sqrt{3}$

$\frac{1+2\sqrt{3}}{2}$

$1 + 2\sqrt{3}$ $\quad$ $\frac{1 + 2\sqrt{3}}{4}$

✓ $\frac{1 + 2\sqrt{3}}{2}$ $\quad$ $\frac{1 + 2\sqrt{3}}{\sqrt{5}}$

Let X be a data matrix of shape (d, n) for a centered dataset. The first principal component of the dataset is $\left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right]^T$. What will the scalar proxy of the point $[1, 2]^T$ be on the first principal component?

Options :

6406531286422. ✖ $1 + 2\sqrt{3}$

6406531286423. ✔ $\frac{1 + 2\sqrt{3}}{2}$

6406531286424. ✖ $\frac{1 + 2\sqrt{3}}{4}$

6406531286425. ✖ $\frac{1 + 2\sqrt{3}}{\sqrt{5}}$

Consider the scenario where you observe that 8 of your friends $\{f_1, \ldots, f_8\}$ have scored the following marks in a quiz: {6, 3, −5, −4, 2, −3, 5, −2} respectively.

Which of the cluster initializations given below will result in clusters where your friends with positive marks are in one cluster and the rest are in another after executing one step of the Lloyd's algorithm?

$I_1 : z_1 = z_2 = z_3 = z_4 = z_5 = z_6 = z_7 = 2$ and $z_8 = 1$

$I_2 : z_1 = 1, z_2 = z_3 = z_4 = z_5 = z_6 = z_7 = z_8 = 2$

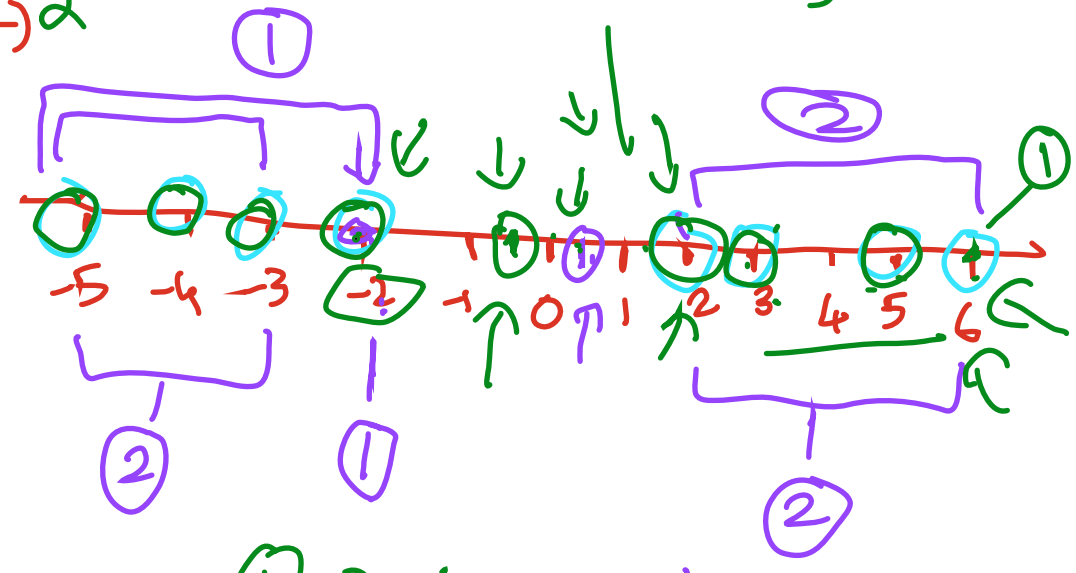
I_1: No, I_2: No

I_1: Yes, I_2: No

I_1: Yes, I_2: Yes

I_1: No, I_2: Yes

Insufficient Information



Consider the scenario where you observe that 8 of your friends $\{f_1, \ldots, f_8\}$ have scored the following marks in a quiz: {6, 3, −5, −4, 2, −3, 5, −2} respectively.

Which of the cluster initializations given below will result in clusters where your friends with positive marks are in one cluster and the rest are in another after executing one step of the Lloyd's algorithm?

$I_1 : z_1 = z_2 = z_3 = z_4 = z_5 = z_6 = z_7 = 2 \text{ and } z_8 = 1$
$I_2 : z_1 = 1, z_2 = z_3 = z_4 = z_5 = z_6 = z_7 = z_8 = 2$

Options :

6406531286429. ✖ I_1: No, I_2: No

6406531286430. ✖ I_1: Yes, I_2: Yes

6406531286431. ✔ I_1: Yes, I_2: No

6406531286432. ✖ I_1: No, I_2: Yes

6406531286433. ✖ Insufficient Information

(0,1.5), (0,0), (0,-1.5)

Consider the following data set:

$$\left\{x_1 = \begin{bmatrix}0\\2\end{bmatrix}, x_2 = \begin{bmatrix}2\\0\end{bmatrix}, x_3 = \begin{bmatrix}0\\0\end{bmatrix}, x_4 = \begin{bmatrix}0\\-2\end{bmatrix}, x_5 = \begin{bmatrix}-2\\0\end{bmatrix}, x_6 = \begin{bmatrix}0\\1\end{bmatrix}, x_7 = \begin{bmatrix}0\\-1\end{bmatrix}\right\}$$

1 2 3

For $k = 3$, assume that the following cluster assignment is given to us which shows the clusters assigned to all data points but x_3 :

$$z = \{1, 2, ?, 3, 2, 1, 3\}$$

What should be the cluster assigned to the data point x_3?

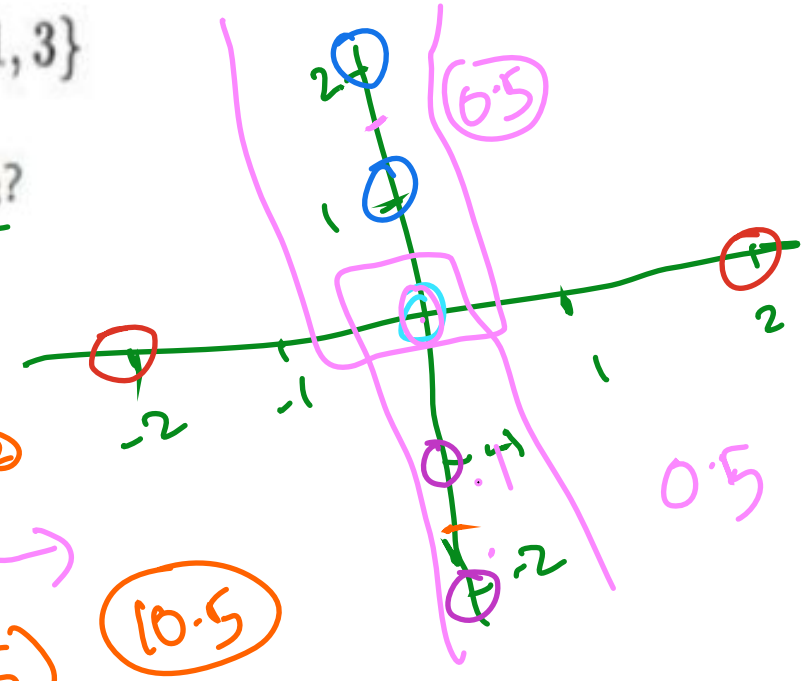
C_1: (0,2), (0,1), (0,0), (0,1)

C_2: (2,0), (-2,0), (0,0), (2,0)

C_3: (0,-2), (0,-1), (0,0), (0,-1.5)

$1^2+0^2+1^2 = 2$

8 (0.5) (0.5)



⇒ 8

9

Consider the following data set:

$$\left\{x_1 = \begin{bmatrix}0\\2\end{bmatrix}, x_2 = \begin{bmatrix}2\\0\end{bmatrix}, x_3 = \begin{bmatrix}0\\0\end{bmatrix}, x_4 = \begin{bmatrix}0\\-2\end{bmatrix}, x_5 = \begin{bmatrix}-2\\0\end{bmatrix}, x_6 = \begin{bmatrix}0\\1\end{bmatrix}, x_7 = \begin{bmatrix}0\\-1\end{bmatrix}\right\}$$

For $k = 3$, assume that the following cluster assignment is given to us which shows the clusters assigned to all data points but x_3 :

$$z = \{1, 2, ?, 3, 2, 1, 3\}$$

What should be the cluster assigned to the data point x_3?

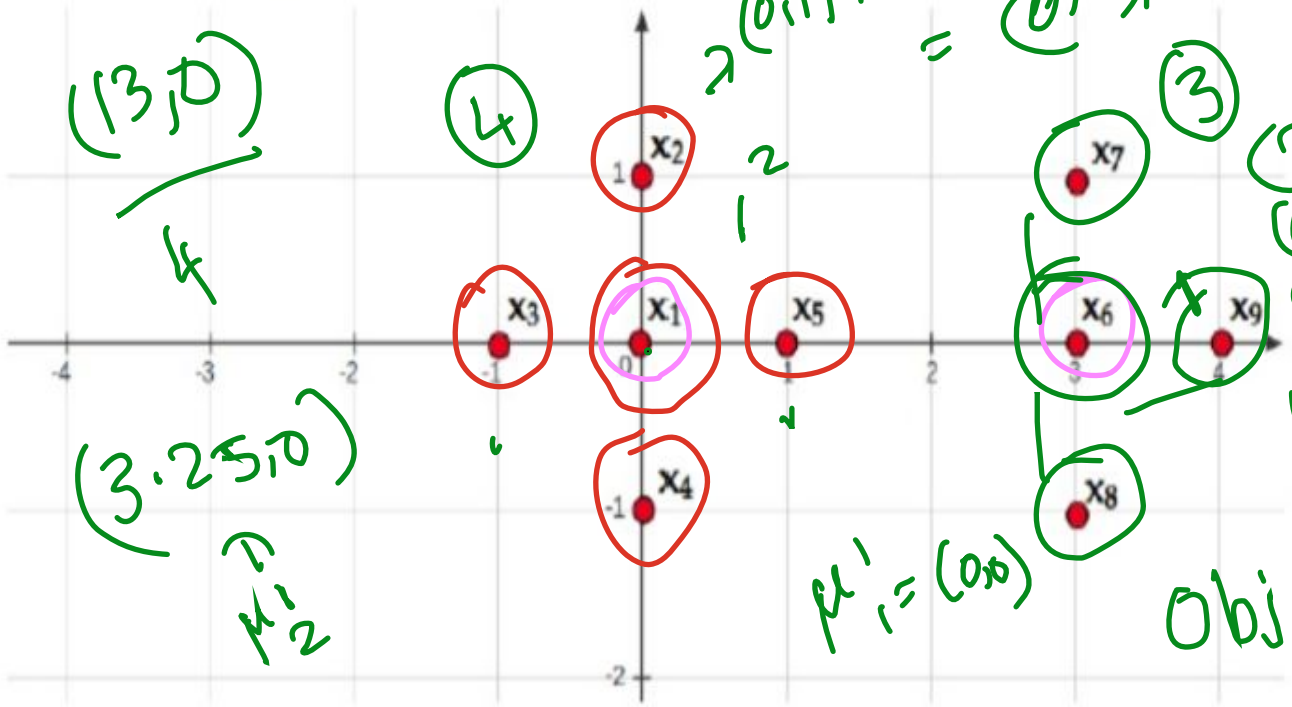
Options :

6406531286426. ✖ 1

6406531286427. ✔ 2

6406531286428. ✖ 3

Consider the data points shown in the following image:



$Obj = 17, \mu_1^1 = (1,0), \mu_2^1 = (3.33,0$

$Obj = 17, \mu_1^1 = (1,0), \mu_2^1 = (3,0)$

$Obj = 7, \mu_1^1 = (0,0), \mu_2^1 = (3.33,0)$

$Obj = 7, \mu_1^1 = (0,0), \mu_2^1 = (3.25,0)$

Consider that in the initialization step of K-means with $k = 2$, the data points x_1 and x_6 got selected as initial cluster centers. That is, $\mu_1^0 = (0,0)$ and $\mu_2^0 = (3,0)$. As per these cluster centers, the data points were then assigned to either cluster 1 or cluster 2. After this assignment, what will be the value of the objective function (Obj)? Further, as per the assignment of data points, the cluster centers will be re-computed for the next iteration. What will be the value of μ_1^1 and μ_2^1?

6406531286434. ✖ $Obj = 17, \mu_1^1 = (1, 0), \mu_2^1 = (3.33, 0)$

6406531286435. ✖ $Obj = 17, \mu_1^1 = (1, 0), \mu_2^1 = (3, 0)$

6406531286436. ✖ $Obj = 7, \mu_1^1 = (0, 0), \mu_2^1 = (3.33, 0)$

6406531286437. ✔ $Obj = 7, \mu_1^1 = (0, 0), \mu_2^1 = (3.25, 0)$

In the context of EM algorithm, select all true statements from the options given below. There are n data-points and K mixtures. The index i corresponds to the ith data-point, the index k corresponds to the kth mixture. f is the density of a Gaussian. All other symbols have their usual meaning.

$\pi_k = P(z_i = k)$ ✓

$\lambda_k^i = P(z_i = k \mid x_i)$ ✓

$\sum_{i=1}^{n} \lambda_k^i = 1$ ✗

$\lambda_k^i = f(x_i \mid z_i = k)$ ✗

$\pi_k = P(z_1 = k)$

$\lambda_k^1 = P(x_i = k \mid x_i)$

1 2 3 4 5

$Z \mid X$

$z_1 = 1.5$

$$\lambda_k^1 = \frac{\pi_k N(x_i, \mu_k, \sigma_k^2)}{\sum_{j=1}^{k} \pi_j N(x_i, \mu_j, \sigma_j^2)}$$

$\lambda_k^1 + \lambda_k^2 + \dots + \lambda_k^n = 1$ ✗

$\lambda_1^1 + \lambda_2^1 + \dots + \lambda_k^1 = 1$

Question Label : Multiple Select Question

In the context of EM algorithm, select all true statements from the options given below. There are n data-points and K mixtures. The index i corresponds to the ith data-point, the index k corresponds to the kth mixture. f is the density of a Gaussian. All other symbols have their usual meaning.

Options :

6406531286438. ✔ $\pi_k = P(z_i = k)$

6406531286439. ✖ $\sum_{i=1}^{n} \lambda_k^i = 1$

6406531286440. ✔ $\lambda_k^i = P(z_i = k \mid x_i)$

6406531286441. ✖ $\lambda_k^i = f(x_i \mid z_i = k)$

Consider a centered dataset of 100 points in $\mathbb{R}^5$. Standard PCA is performed on this dataset. Consider the following ratio:

$$\theta = \left(\frac{\text{variance along the first principal component}}{\text{sum of variances along all five principal components}} \right)$$

What is the minimum value of θ?

$$\frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5}$$

$$\frac{\lambda_1}{5\lambda_1} = \frac{1}{5}$$

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq \lambda_5 \geq 0$$

Consider a centered dataset of 100 points in $\mathbb{R}^5$. Standard PCA is performed on this dataset. Consider the following ratio:

$$\theta = \frac{\text{variance along the first principal component}}{\text{sum of variances along all five principal components}}$$

What is the minimum value of θ?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.19 to 0.21

Consider a centered dataset of 1000 points in $\mathbb{R}^{10}$. Standard PCA is performed on this dataset. What is the maximum number of principal components that can be retained so that the compression ratio does not fall below 1.4 ? The compression ratio is defined as:

$$\frac{\text{size of original dataset}}{\text{size of reconstructed dataset}} \geq 1.4$$

$k \leq \frac{10000}{101 \times 14}$

1010×1.4

101×14

$\frac{10000}{1010k} \geq 1.4$

$\frac{10000}{101 \times 14} \geq k$

Assume that each real number occupies one unit of storage space. Note that we are interested in the reconstructions and not just in the scalar projections. So, the reconstructed dataset will also be in $\mathbb{R}^{10}$.

$\boxed{10k}$

$\mathbb{R}^{10} \rightarrow \underbrace{w_1, \ldots, w_{10}}_{k \rightarrow}$

$w \in \mathbb{R}^{10}$

$7.0721 \geq k \rightarrow 7$

$w \in \mathbb{R}^{10}$

$w_1, \ldots, w_k$

$$\underline{x_1} = \underbrace{(x_1^T w_1)}_{\mathbb{R}} \underline{w_1} + \cdots + \underbrace{(x_1^T w_k)}_{\mathbb{R}} \underline{w_k} \Rightarrow \mathbb{R}$$

$\boxed{1000k}$

$\boxed{10k}$

$\boxed{1010k}$

Consider a centered dataset of 1000 points in $\mathbb{R}^{10}$. Standard PCA is performed on this dataset. What is the maximum number of principal components that can be retained so that the compression ratio does not fall below 1.4 ? The compression ratio is defined as:

$$\frac{\text{size of original dataset}}{\text{size of reconstructed dataset}}$$

Assume that each real number occupies one unit of storage space. Note that we are interested in the reconstructions and not just in the scalar projections. So, the reconstructed dataset will also be in $\mathbb{R}^{10}$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

7 ✓

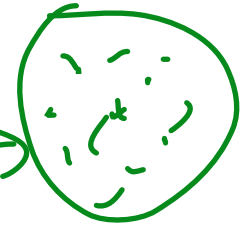
$B_p \alpha$ ~~B_p~~ ~~B_p~~ p

Consider a dataset with 60 points in which all points are either 0 or 1. We use a Bernoulli distribution with parameter p to model this problem. The prior and posterior distributions for the parameter p are $\text{Beta}(10, 5)$ and $\text{Beta}(30, 45)$ respectively. How many data-points have the value 0 in this dataset? Note that $p = P(x = 1)$ as usual.

$\text{Beta}(10,5)$

$n_0 = 40$

$n_1 = 20$

Bernoulli(p) →  → $\text{Beta}(30, 45)$

$\frac{\alpha}{\alpha+\beta}$

Prior α, β → Posterior mean

$$\frac{n_0 + 10}{60 + 10 + 5} = \frac{n_0 + \alpha}{n + \alpha + \beta} = \frac{30}{30 + 45}$$

$$\frac{n_0 + 10}{75} = \frac{30}{75}$$

Consider a dataset with 60 points in which all points are either 0 or 1. We use a Bernoulli distribution with parameter p to model this problem. The prior and posterior distributions for the parameter p are $\mathrm{Beta}(10, 5)$ and $\mathrm{Beta}(30, 45)$ respectively. How many data-points have the value 0 in this dataset? Note that $p = P(x = 1)$ as usual.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

40

A GMM is fit for a dataset with 5 points. At some time-step in the EM algorithm, the following are the values of λ_k^i for all points in the dataset for the kth mixture after the E-step:

$$\lambda_k^1 = 0.9$$
$$\lambda_k^2 = 0.8$$
$$\lambda_k^3 = 0.7$$
$$\lambda_k^4 = 0.6$$
$$\lambda_k^5 = 0.5$$

What is the estimate of π_k after the M-step?

$$\pi_k = \frac{\sum_{i=1}^{n} \lambda_k^i}{n} = \frac{3.5}{5} = 0.7$$

A GMM is fit for a dataset with 5 points. At some time-step in the EM algorithm, the following are the values of λ_k^i for all points in the dataset for the kth mixture after the E-step:

$$\lambda_k^1 = 0.9$$
$$\lambda_k^2 = 0.8$$
$$\lambda_k^3 = 0.7$$
$$\lambda_k^4 = 0.6$$
$$\lambda_k^5 = 0.5$$

What is the estimate of π_k after the M-step?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.69 to 0.71

$\mathbb{R}^2 \to \mathbb{R}^6$

Consider the dataset as shown in the figure. We want to project the dataset into another feature space so that it lives in a linear subspace of a higher-dimension space. We apply the kernel PCA with a polynomial kernel of the appropriate degree to achieve the same. What will be the dimension of transformed feature space?

$(\vec{x}^T \vec{y} + c)^2$

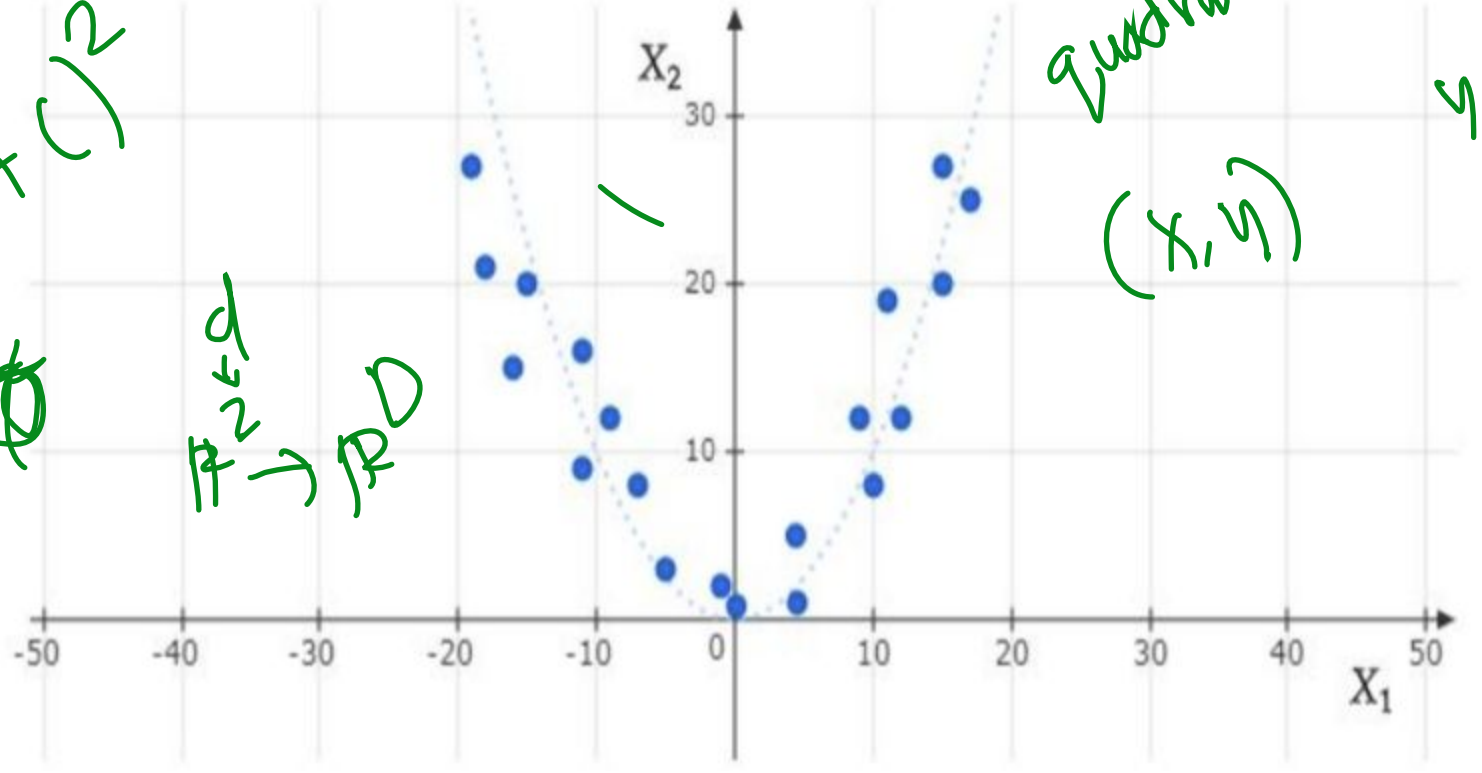
$\mathbb{R}^{2+d} \to \mathbb{R}^D$

$\mathbb{R}$ $\mathbb{P}$ ϕ

quadratic

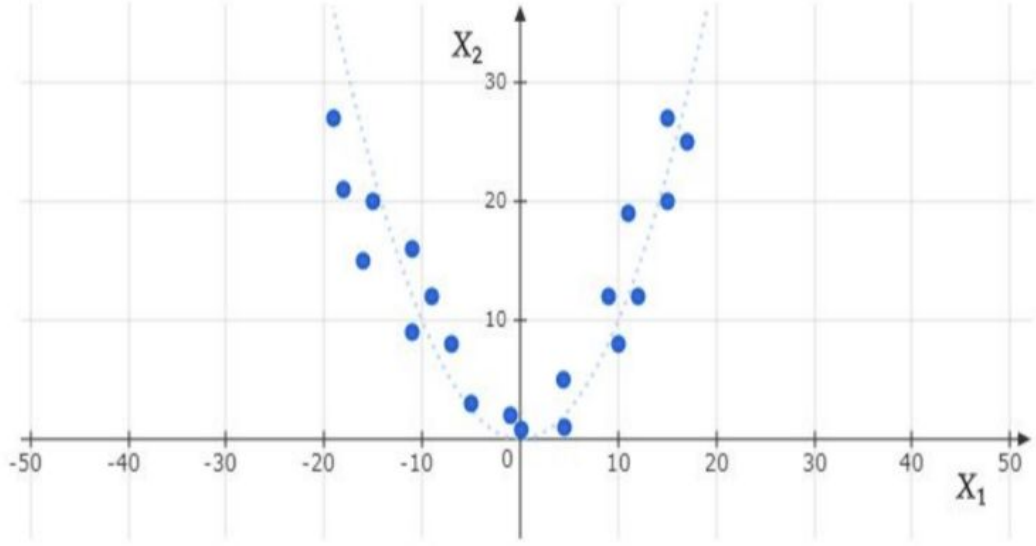
(x, y)

$y = \text{quad}(x)$



$$2+2 \quad C_2 \qquad 4C_2 \qquad \frac{4!}{2!\,2!} \qquad \frac{24}{2\times 2} = \frac{24}{4} = 6$$

Consider the dataset as shown in the figure. We want to project the dataset into another feature space so that it lives in a linear subspace of a higher-dimension space. We apply the kernel PCA with a polynomial kernel of the appropriate degree to achieve the same. What will be the dimension of transformed feature space?



Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

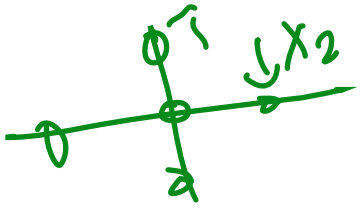
Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Consider the following data set:



$$\left\{x_1 = \begin{bmatrix}0\\2\end{bmatrix}, x_2 = \begin{bmatrix}2\\0\end{bmatrix}, x_3 = \begin{bmatrix}0\\0\end{bmatrix}, x_4 = \begin{bmatrix}0\\-2\end{bmatrix}, x_5 = \begin{bmatrix}-2\\0\end{bmatrix}\right\}$$

With $k = 2$, as per k-means++,

What is the probability of the following points x_2, x_1 (in that order) being chosen as initial cluster centers?

$1^{st} \Rightarrow$ randomly

$$P(x_2) = \frac{1}{n} = \frac{1}{5} = 0.2$$

$$P(\mu_1^0 = x_2) = 0.2$$

$$P(\mu_2^0 = x_1) = \frac{S(x_1)}{\sum S(x_i) \forall i}$$

What is the probability of the following points x_2, x_1 (in that order) being chosen as initial cluster centers?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

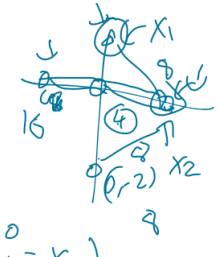
0.03 to 0.05

Consider the following data set:

$$\left\{ x_1 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, x_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, x_4 = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, x_5 = \begin{bmatrix} -2 \\ 0 \end{bmatrix} \right\}$$

With $k = 2$, as per k-means++,

What is the probability of the following points x_2, x_3 (in that order) being chosen as initial cluster centers?



$$\left\| \begin{pmatrix} 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right\|^2 = \left\| \begin{pmatrix} -2 \\ 2 \end{pmatrix} \right\|^2 \quad 8$$

$$P(\mu_1^0 = x_2) = 0.2 \qquad \frac{8}{8+8+4+16}$$

$$\frac{8}{36} \qquad \frac{2}{9} = 0.44$$

$$P(\mu_2^0 = x_1) = \frac{2}{9}$$

$$\frac{4}{36} = \frac{1}{9}$$

$$\frac{1}{5} \times \frac{2}{9} = \frac{2}{45} \rightarrow P(\mu_1^0 = x_2, \mu_2^0 = x_1)$$

$$\frac{1}{5} \times \frac{1}{9} = \frac{1}{45}$$

What is the probability of the following points x_2, x_3 (in that order) being chosen as initial cluster centers?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.01 to 0.03

Consider the following data set:

$$\left\{x_1 = \begin{bmatrix}0\\2\end{bmatrix}, x_2 = \begin{bmatrix}2\\0\end{bmatrix}, x_3 = \begin{bmatrix}0\\0\end{bmatrix}, x_4 = \begin{bmatrix}0\\-2\end{bmatrix}, x_5 = \begin{bmatrix}-2\\0\end{bmatrix}\right\}$$

With $k = 2$, as per k-means++,

What is the probability of the following points x_2, x_5 (in that order) being chosen as initial cluster centers?

What is the probability of the following points x_2, x_5 (in that order) being chosen as initial cluster centers?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.08 to 0.095

Consider a dataset that has 4 points in $\mathbb{R}^2$ that lie on a line passing through the origin:

$$D = \left\{ \begin{bmatrix} -2 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\} \begin{matrix} x \\ y \end{matrix}$$

$y = 3x$

$\frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$\frac{1}{\sqrt{10}}$

Based on the above data, answer the given subquestions.

Is this dataset centered?

Yes

No

Choose all representatives $\mathbf{w}$ for this dataset such that $||\mathbf{w}|| = 1$.

$$\frac{1}{\sqrt{10}} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ -3 \end{bmatrix}$$

$$\frac{1}{\sqrt{10}} \cdot \begin{bmatrix} -1 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

Is this dataset centered?

Options :

6406531286450. ✔ Yes

6406531286451. ✖ No

Choose all representatives $\mathbf{w}$ for this dataset such that $||\mathbf{w}|| = 1$.

Options :

6406531286452. ✔ $\frac{1}{\sqrt{10}} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

6406531286453. ✔ $\frac{1}{\sqrt{10}} \cdot \begin{bmatrix} -1 \\ -3 \end{bmatrix}$

6406531286454. ✖ $\begin{bmatrix} -1 \\ -3 \end{bmatrix}$

6406531286455. ✖ $\begin{bmatrix} 2 \\ 6 \end{bmatrix}$

Consider a dataset that has 4 points in $\mathbb{R}^2$ that lie on a line passing through the origin:

$$D = \left\{ \begin{bmatrix} -2 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\}$$

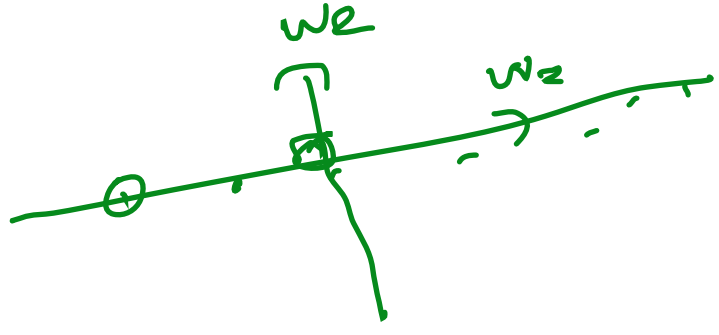
40 10 10 40

$$\frac{100}{n} = \frac{\sum_{i=1}^{n} (x_i^T w_1)}{n}$$

Based on the above data, answer the given subquestions.

If standard PCA is performed on this dataset, what is the variance along the first principal component? 25

If standard PCA is performed on this dataset, what is the variance along the second princip component? 0



If standard PCA is performed on this dataset, what is the variance along the first principal component?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

25 ✓

If standard PCA is performed on this dataset, what is the variance along the second principal component?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0 ✓

Assume that you have a dataset of four points $\{x_1, x_2, x_3, x_4\}$, all of which lie in $[0, 1]$. You hypothesise that the data points are iid random variables with the following density:

$$f(x;\theta) = \begin{cases} \theta x^{\theta-1}, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$\theta = \frac{-4}{\ln(x_1 \cdots x_4)}$

$a^m \quad b^m$

$(ab)^m$

$L = -\theta \ln(x_1 \cdots x_4)$

$\prod_{i=1}^{4} f(x_i|\theta)$

$\log_e\left(\theta^4 (x_1 \cdots x_4)^{(\theta-1)}\right)$

Based on the above data, answer the given subquestions.

What is the log-likelihood of this dataset under this distribution?

You are given the actual values of these observations:

$$x_1 = \frac{1}{e}, \quad x_2 = \frac{1}{e^2}, \quad x_3 = \frac{1}{e^3}, \quad x_4 = \frac{1}{e^4}$$

$\rightarrow \frac{1}{e^{10}}$

$\frac{4}{\theta} + \ln(x_1 \cdots x_4) = 0$

$\log_e(\theta^4)$

What is the maximum likelihood estimate for θ? Use $\log_e$ wherever appropriate.

Hint: Differentiate the log-likelihood with respect to θ and set it to zero.

$4 \ln \theta + (\theta - 1) \ln(x_1 \cdots x_4)$

$$\frac{-4}{\ln\left(e^{-10}\right)}$$

$$\frac{-4}{-10} = 0.4$$

$$\boxed{\theta = 0.4}$$

What is the log-likelihood of this dataset under this distribution?

6406531286458. ✖ $\prod_{i=1}^{4} \theta x_i^{\theta-1}$

6406531286460. ✔ $\sum_{i=1}^{4} [\log \theta + (\theta - 1) \log x_i]$

6406531286459. ✖ $\sum_{i=1}^{4} \theta x_i^{\theta-1}$

6406531286461. ✖ $\prod_{i=1}^{4} [\log \theta + (\theta - 1) \log x_i]$

Question Label : Short Answer Question

You are given the actual values of these observations:

$$x_1 = \frac{1}{e}, \quad x_2 = \frac{1}{e^2}, \quad x_3 = \frac{1}{e^3}, \quad x_4 = \frac{1}{e^4}$$

What is the maximum likelihood estimate for θ? Use $\log_e$ wherever appropriate.

Hint: Differentiate the log-likelihood with respect to θ and set it to zero.

Possible Answers :

0.39 to 0.41